

Task 3: Write a short note on S3. What is difference between Static website and Dynamic website?

Create a static website using S3 with minimum 3 pages having images and videos.

- **A short note on S3:** - Amazon S3, or Simple Storage Service, is a cloud-based storage service provided by Amazon Web Services (AWS).
 - **AWS S3** offers scalable object storage for data storage and retrieval.
 - AWS S3 allows you to store and retrieve any amount of data from anywhere on the web, providing a highly available and durable storage infrastructure.
 - It's simple and cost-effective, with features like versioning, encryption, and lifecycle management for data governance and security.
 - Where Aws S3 is widely used for various purposes, including backup and recovery, data archiving, website hosting, and data analytics. With its pay-as-you-go pricing model.
-
- **Difference between Static website and Dynamic website:** -

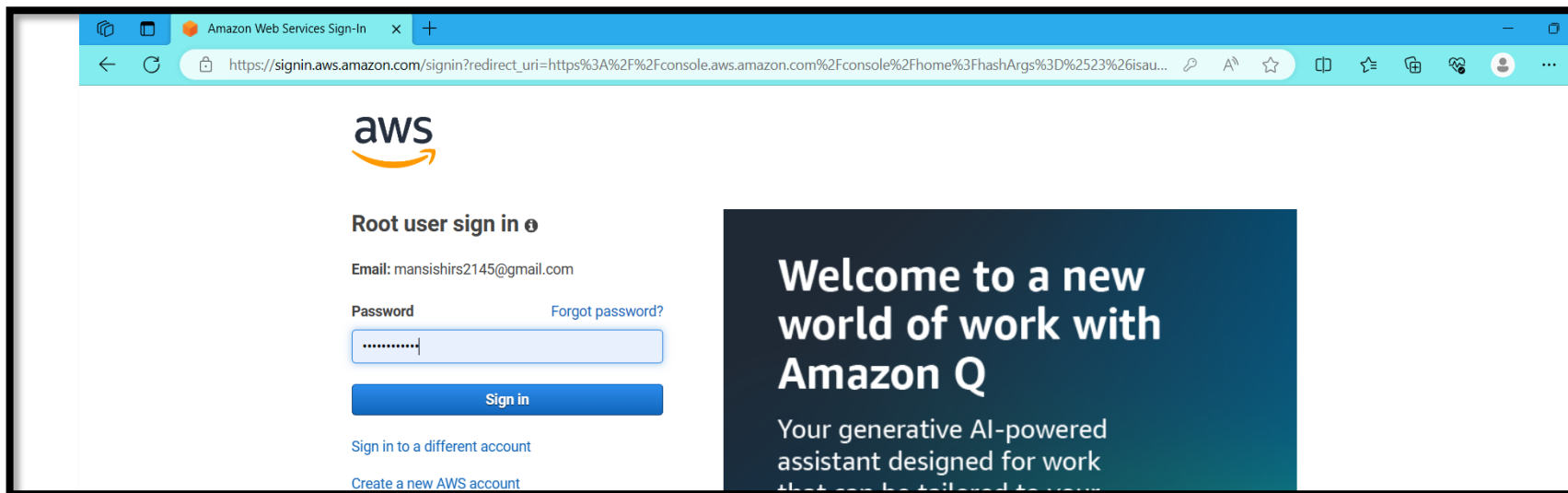
Static website		Dynamic website	
1	A static website in AWS S3 is a collection of HTML, CSS, JavaScript, and other static files stored in an S3 bucket.	1	A dynamic website refers to a web application or site that generates content dynamically based on user interactions or requests.
2	The files are made publicly accessible, and the S3 bucket is configured to function as a web server, delivering content directly to users when they access the provided website URL.	2	Unlike static websites, which serve pre-defined content directly from S3 buckets, dynamic websites involve server-side processing to generate web pages on-the-fly in response to user actions.
3	A static website hosted on AWS S3 refers to a website that consists of fixed, unchanging content delivered directly to the user's web browser without any server-side processing.	3	To host a dynamic website on AWS, you typically use services like AWS Lambda, Amazon API Gateway, Amazon DynamoDB, or AWS Elastic Beanstalk in conjunction with S3.

4	users can enable static website hosting for a bucket, configuring it to serve content directly to web browsers	4	These services handle the server-side processing, database interactions, and other dynamic functionalities required by the website.
5	AWS S3, a scalable object storage service, provides a cost-effective and straightforward way to host static websites.	5	It offers the benefits of scalability, cost-effectiveness, and ease of management associated with serverless and cloud-native architectures.

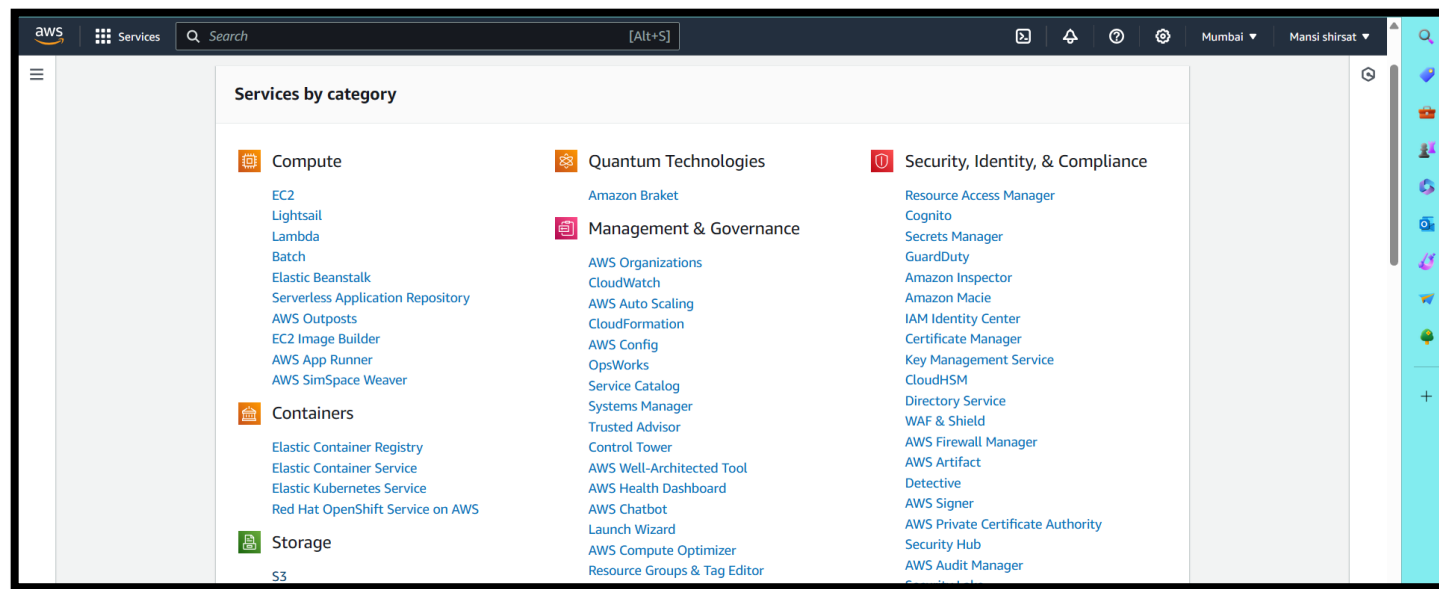
Create a static website using S3 with minimum 3 pages having images and videos.

➤ *Steps for creating Static Website using S3*

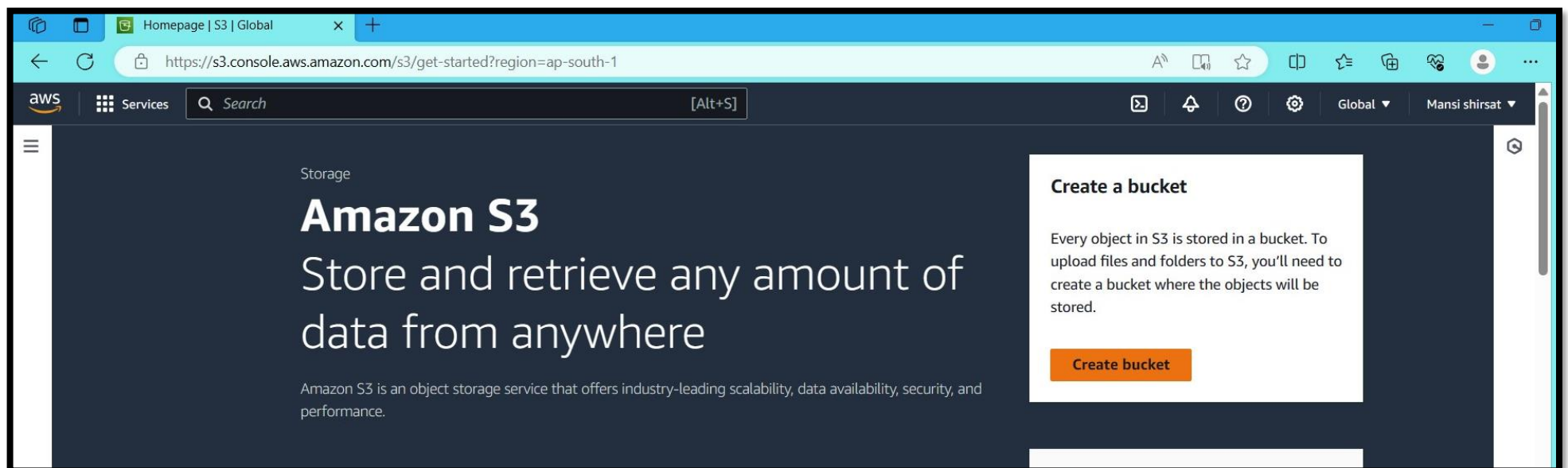
1. **Sign in to AWS Management Console:** -Log in to the AWS Management Console using your AWS account credentials.



2. **Open the Amazon S3 Console:** - Navigate to the Amazon S3 console/ Search S3 in search bar.
WE can see in all services in storage S3.



3. **Create a Bucket:** - *Click on "Create bucket."
- *Enter a unique bucket name and choose a region.
- *Click "Create."



aws

Services

Search

[Alt+S]

Global

Mansi shirsat

Amazon S3

Buckets

Create bucket

Create bucket

Info

Buckets are containers for data stored in S3.

General configuration

AWS Region

Asia Pacific (Mumbai) ap-south-1

Bucket name

Info

mystaticwebsitesbucket

Bucket name must be unique within the global namespace and follow the bucket naming rules. [See rules for bucket naming](#)

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

aws

Services

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Global

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Amazon S3

Buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Object Ownership

Info

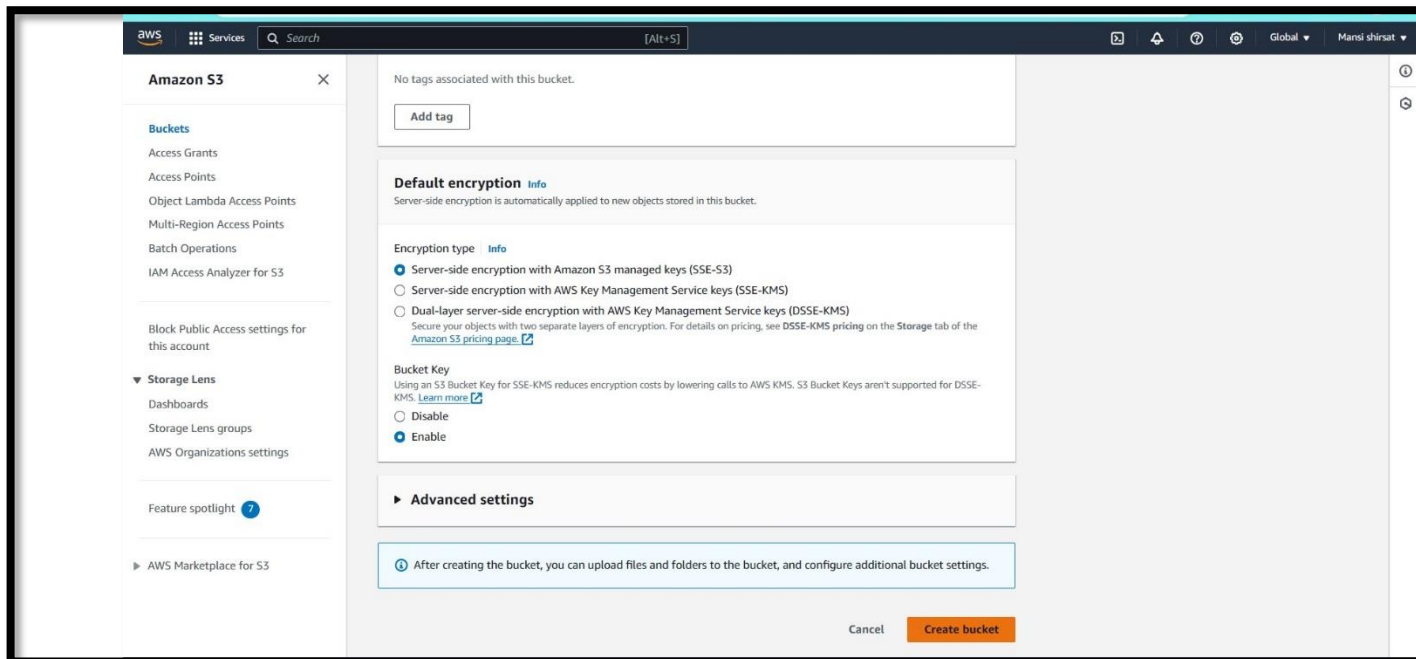
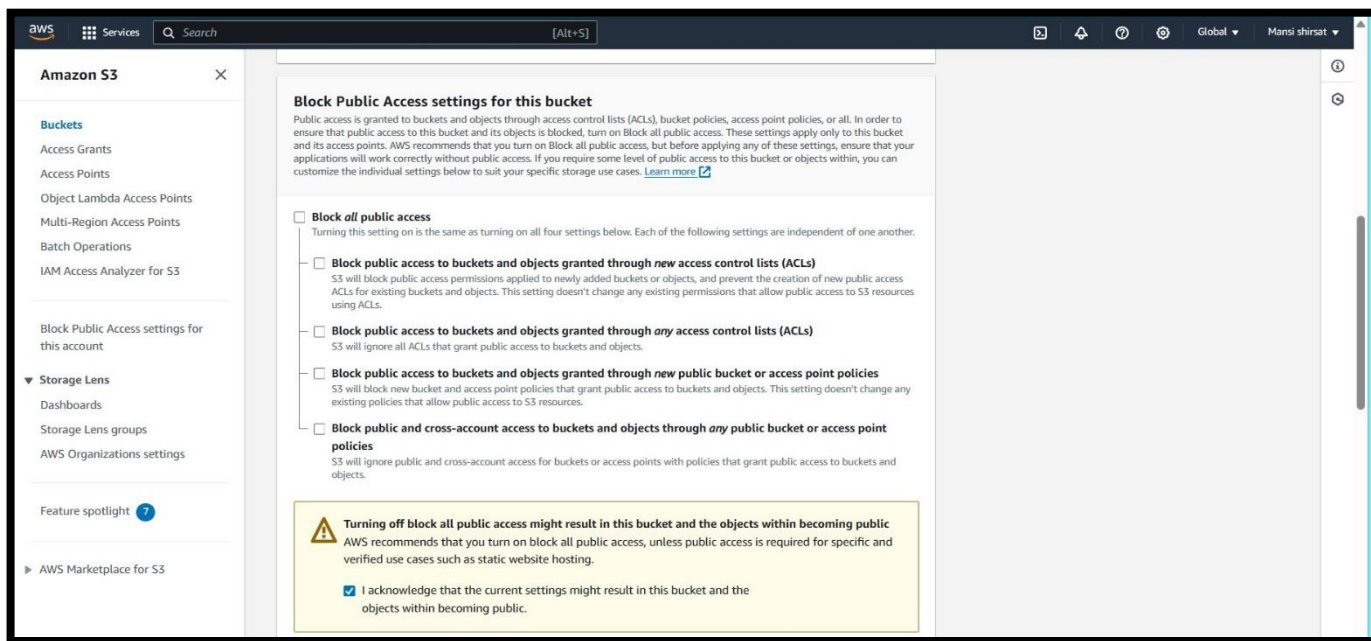
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

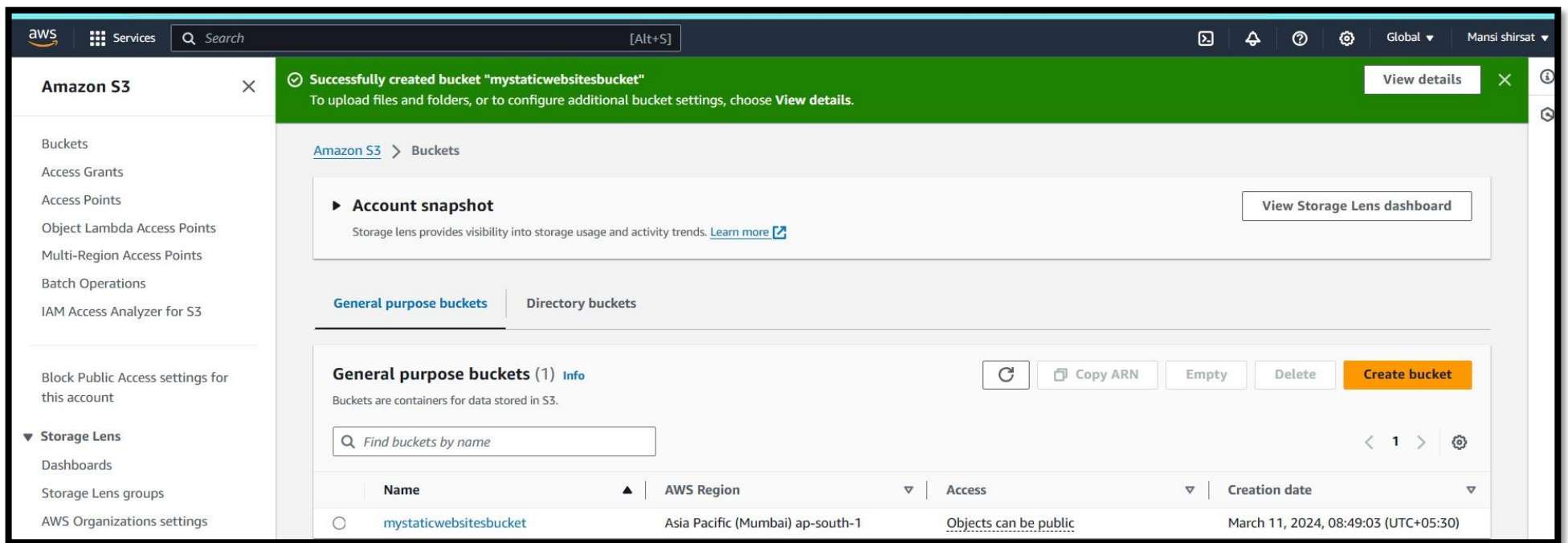
☐ ACLs disabled (recommended)

All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☒ ACLs enabled

Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.





4. Enable Static Website Hosting: - *Select the newly created bucket.

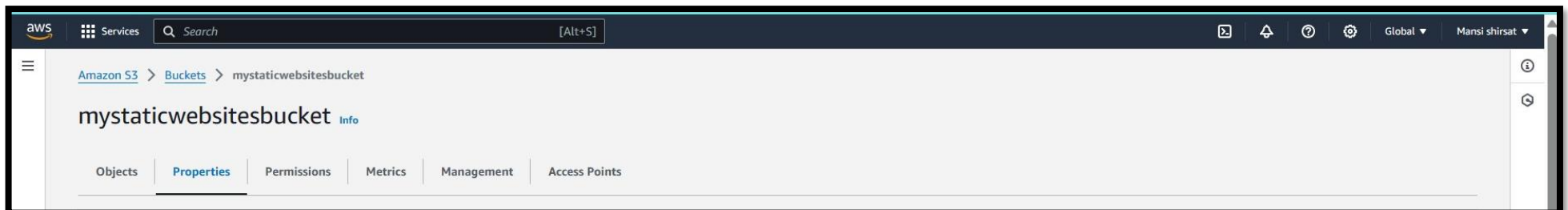
*Go to the "Properties" tab.

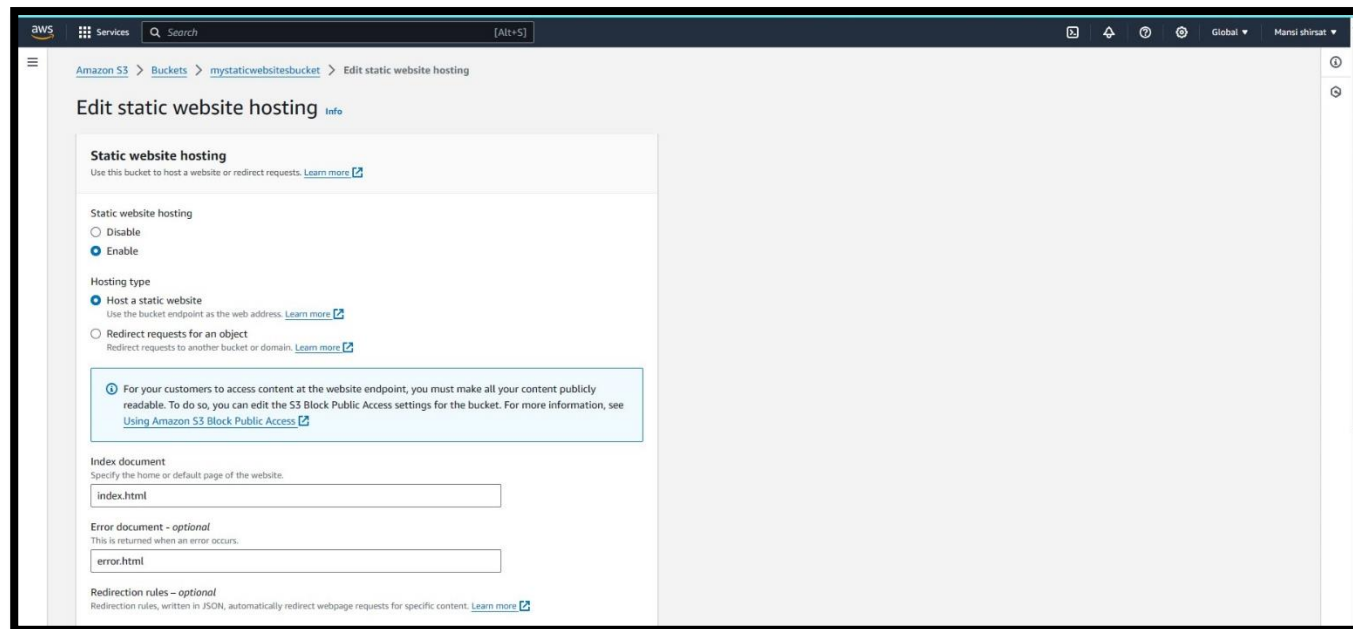
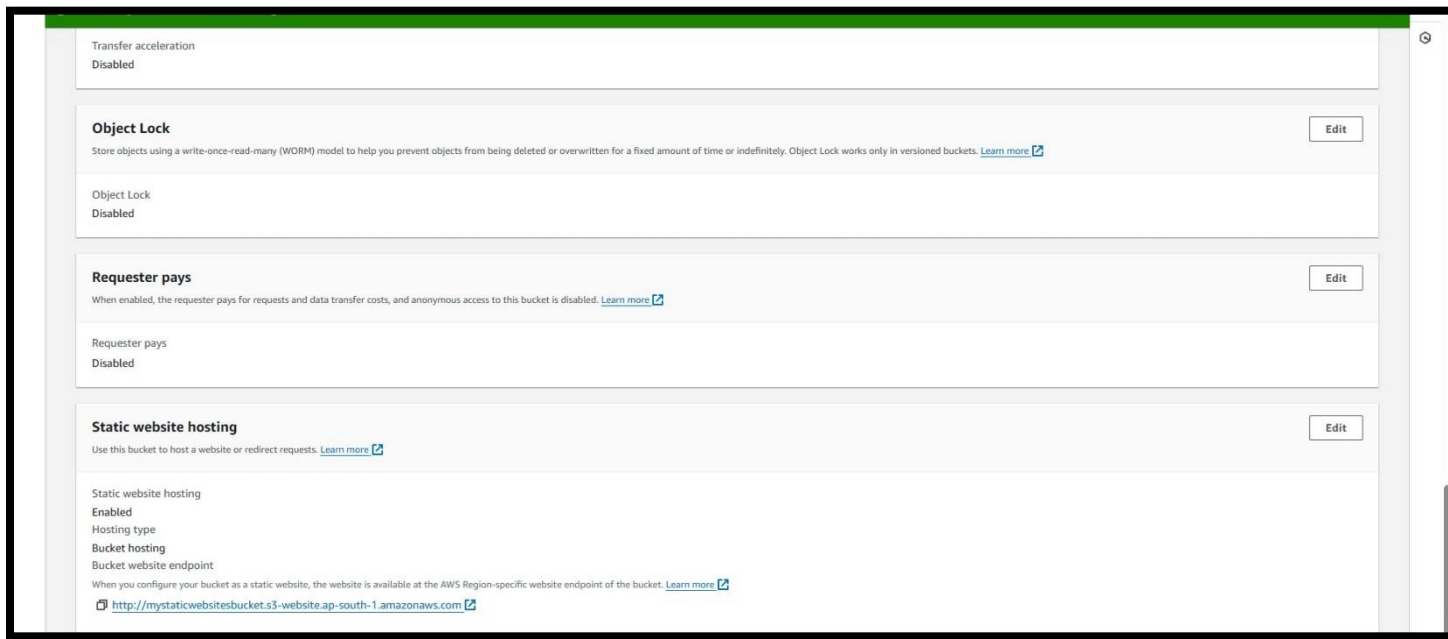
*Click on "Static website hosting."

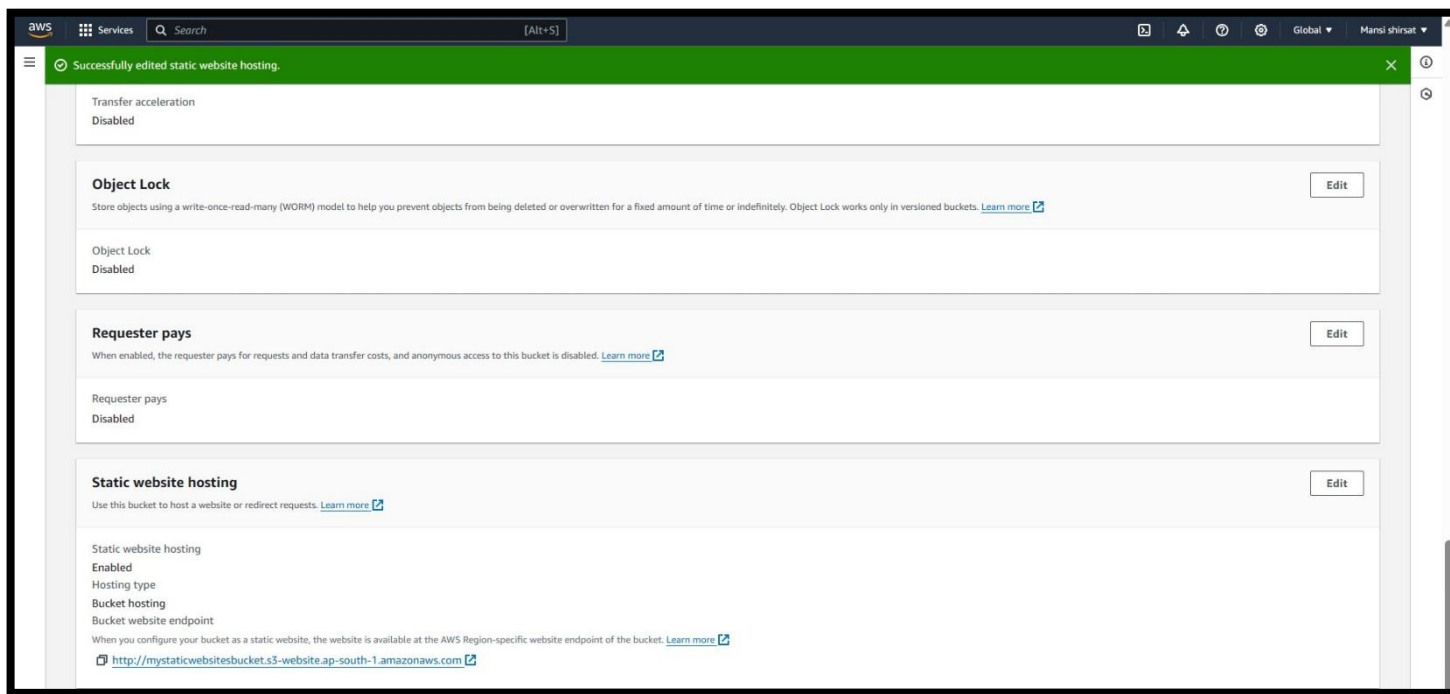
*Choose "Use this bucket to host a website."

*Enter the names of your index and error documents (e.g., index.html, error.html).

*Click "Save."



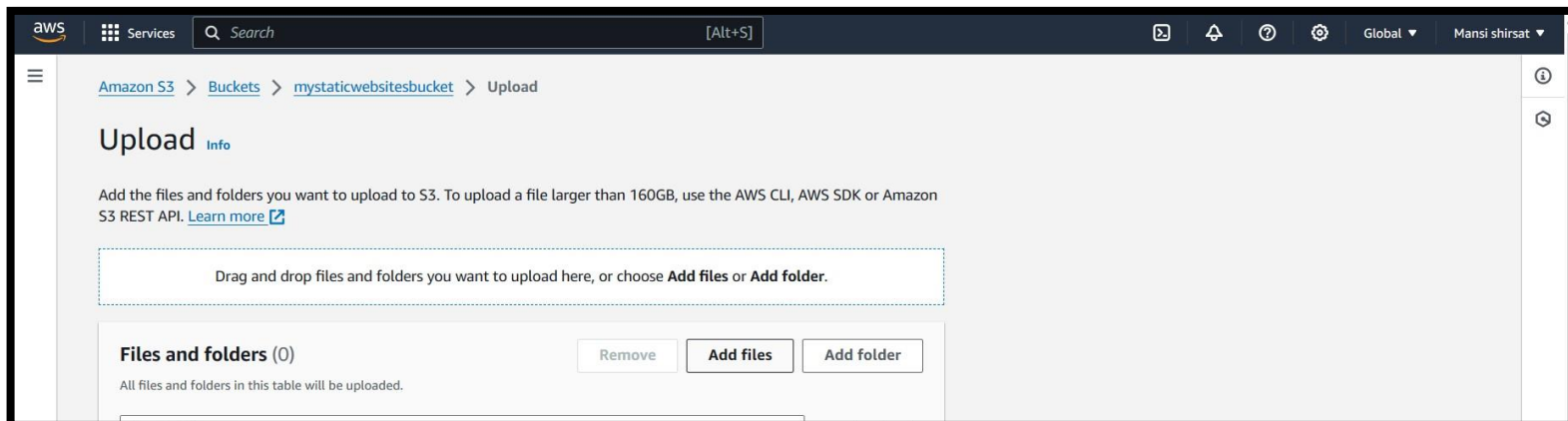


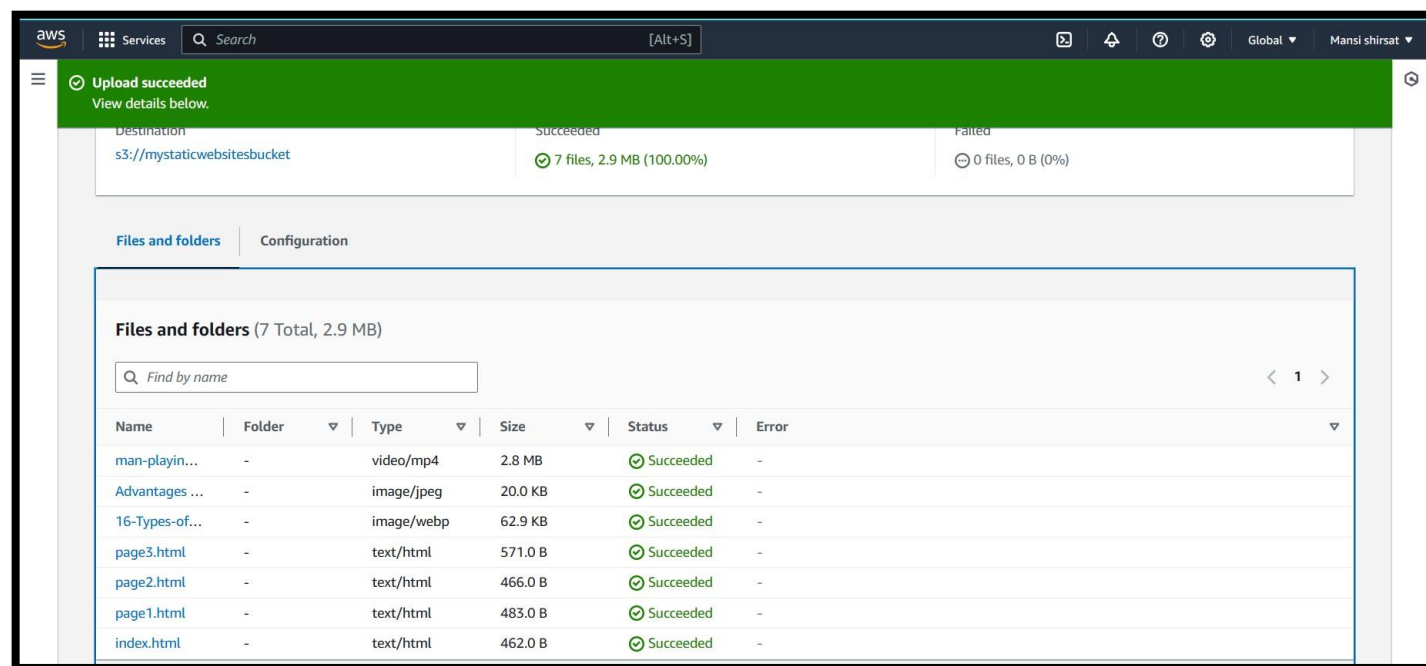


5. Upload Content: - *Go to the "Overview" tab.

***Click "Upload" to upload your HTML, CSS, images, and video files.**

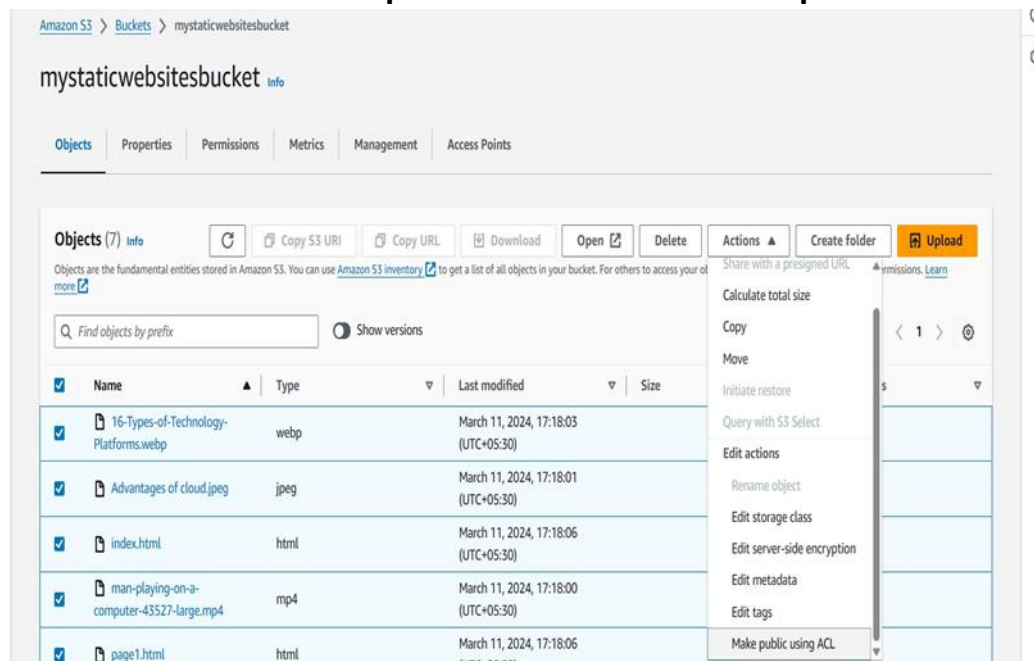
***Click "Add files" and select your files.**

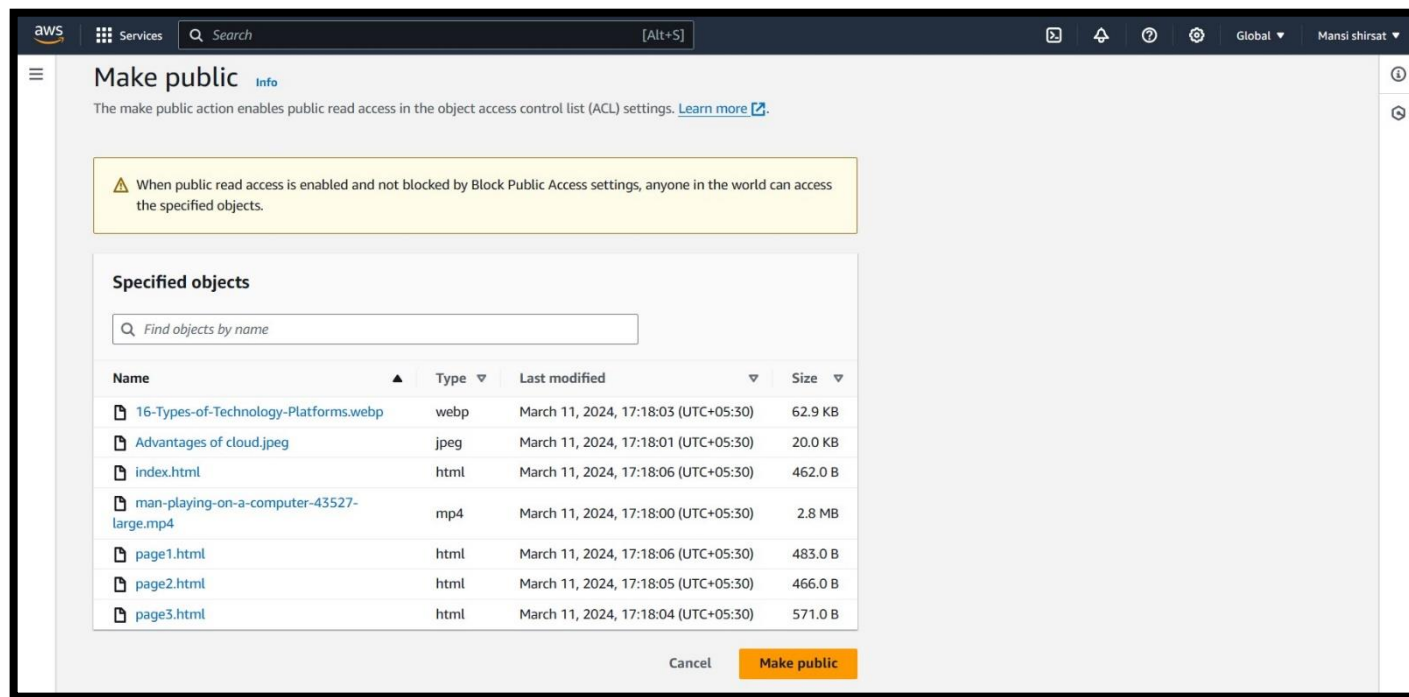




6. Set Permissions: - *Select the uploaded files.

*Go to the "Actions" dropdown and choose "Make public" to ensure public access.





7. Create HTML Pages: - *Create at least three HTML pages (e.g., index.html, page1.html, page2.html) with links to images and videos.

*Add content, images, and video tags as needed.

*Open Notepad

***Type the below code: for index.html**

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Static Website</title>
</head>
<body>
  <h1>Welcome to My Static Website</h1>
  <p>Click below to explore:</p>
  <ul>
    <li><a href="page1.html">Page 1</a></li>
```

```
    <li><a href="page2.html">Page 2</a></li>
</li><a href="page3.html">Page 3</a></li>
</ul>
</body>
</html>
```

Then create new page1.html page

```
Code: -<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Page 1</title>
</head>
<body>
  <h1>Welcome to Page 1</h1>

  <h2>Image:</h2>
  

  <p><a href="page2.html">Go to Page 2</a></p>
  <p><a href="page3.html">Go to Page 3</a></p>
</body>
</html>
```

- create new page2.html page

```
code: - <!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Page 1</title>
</head>
<body>
```

```
<h1>Welcome to Page 2</h1>
```

```
<h2>Image:</h2>
```

```

```

```
<p><a href="page1.html">Go to Page 1</a></p>
```

```
<p><a href="page3.html">Go to Page 3</a></p>
```

```
</body>
```

```
</html>
```

- **create new page3.html page**

```
code: - <!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Page 3</title>
```

```
</head>
```

```
<body>
```

```
  <h1>Welcome to Page 3</h1>
```

```
  <h2>Video:</h2>
```

```
  <video width="320" height="240" controls>
```

```
    <source src="https://mystaticwebsitesbucket.s3.ap-south-1.amazonaws.com/man-playing-on-a-computer-43527-large.mp4"
type="video/mp4">
```

```
  </video>
```

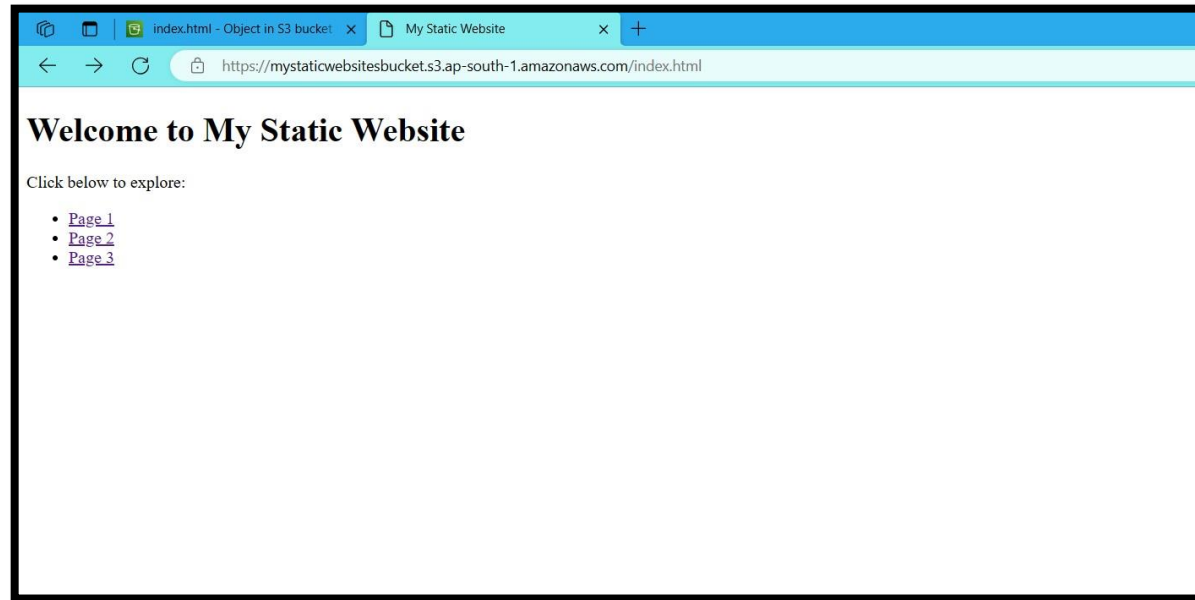
```
  <p><a href="page1.html">Go to Page 1</a></p>
```

```
  <p><a href="page2.html">Go to Page 2</a></p>
```

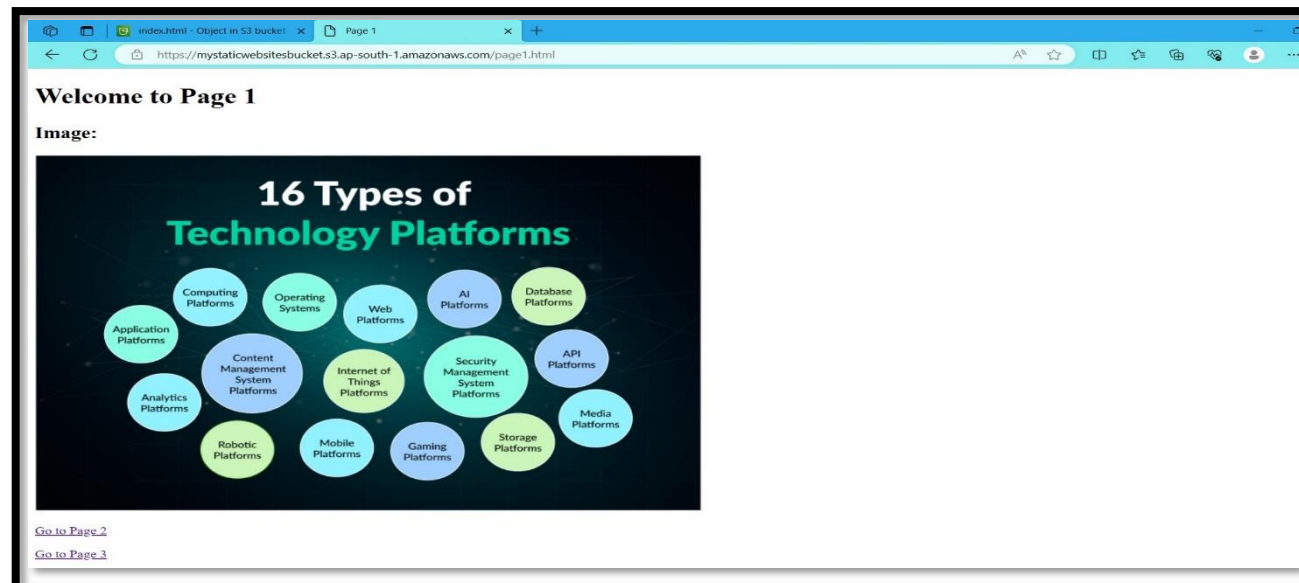
```
</body>
```

```
</html>
```

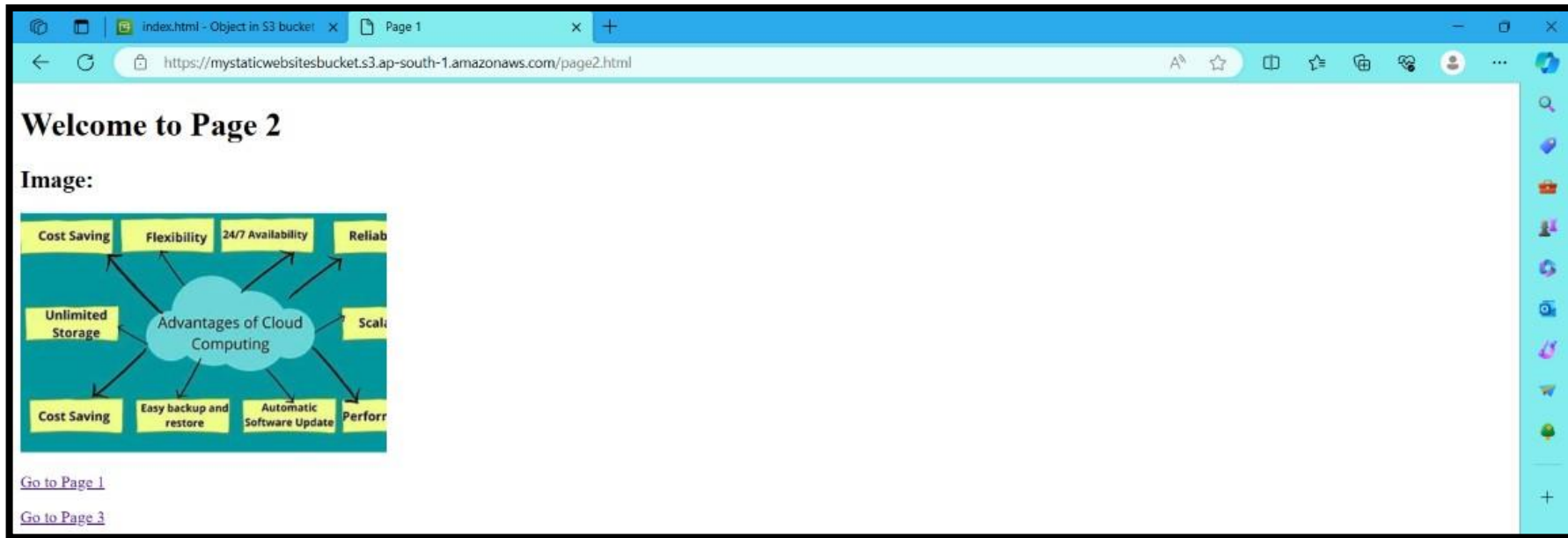
8. Testing: - *Use the static website endpoint provided in the "Static website hosting" section to access your website.
By copying url from s3 object url :- <https://mystaticwebsitesbucket.s3.ap-south-1.amazonaws.com/index.html>
& paste in browser url. You can see outputs below: -



when you click on page 1 you see page1 output below



when you click on page2 you see page2 output below



when you click on page3 you see page3 output below

